

三. 二维共形不变性

1. 二维共形群

2维坐标 (z^0, z^1) $z^\mu \rightarrow w^\mu(x)$, $g^{\mu\nu} \rightarrow (\frac{\partial w^\mu}{\partial z^\alpha})(\frac{\partial w^\nu}{\partial z^\beta})g^{\alpha\beta}$

共形变换: $g'_{\mu\nu}(w) \propto g_{\mu\nu}(z)$, 使变换满足: $(\frac{\partial w^0}{\partial z^0})^2 + (\frac{\partial w^0}{\partial z^1})^2 = (\frac{\partial w^1}{\partial z^0})^2 + (\frac{\partial w^1}{\partial z^1})^2$; $\frac{\partial w^0}{\partial z^0} \frac{\partial w^1}{\partial z^0} + \frac{\partial w^0}{\partial z^1} \frac{\partial w^1}{\partial z^1} = 0$

上式即: $\frac{\partial w^1}{\partial z^0} = \frac{\partial w^0}{\partial z^1}$ 且 $\frac{\partial w^0}{\partial z^0} = -\frac{\partial w^1}{\partial z^1}$

可同构到复平面: $z = z^0 + iz^1$, $\bar{z} = z^0 - iz^1 \Rightarrow$ 记 ∂ 为 ∂_z , $\bar{\partial}$ 为 $\partial_{\bar{z}}$

$g^{\mu\nu} = (\frac{1}{2} \frac{1}{2})$, $g_{\mu\nu} = (2^2)$, 反对称度规 $\epsilon_{\mu\nu} = (\frac{1}{2} -\frac{1}{2})$, $\epsilon^{\mu\nu} = (2; -2i)$

复函数 $\partial_{\bar{z}} W(z, \bar{z}) = 0 \rightarrow$ 解为 $W(z) \rightarrow dW = (\frac{dW}{dz})dz$

$z^* = \bar{z}$ 为物理域

可逆变换 $\rightarrow$ 全局

因(荷)

$\hookrightarrow f(z) = \frac{az+b}{cz+d}$, $ad-bc=1$ 射影变换, 特征矩 $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$f_1 \circ f_2 = \frac{a_1 \frac{a_2 z + b_2}{c_2 z + d_2} + b_1}{c_1 \frac{a_2 z + b_2}{c_2 z + d_2} + d_1} = \frac{(a_1 a_2 + b_1 c_2)z + (a_1 b_2 + b_1 d_2)}{(c_1 a_2 + d_1 c_2)z + c_1 d_2 + d_1 b_2} \Rightarrow A' = A_2 A_1$

2d全局共形对称 $\sim SL(2, \mathbb{C}) \sim SO(1, 3)$

局域对称性: $z' = z + \epsilon(z) \Rightarrow \epsilon(z) = \sum_{n=1}^{\infty} c_n z^n$ 洛朗展开 $z=0$

$\downarrow$

无旋无维场 $\phi'(z, \bar{z}) = \phi(z', \bar{z}') - \epsilon(z')\partial\phi(z, \bar{z}') - \bar{\epsilon}(\bar{z}')\bar{\partial}\phi(z', \bar{z}')$

$\delta\phi = -\epsilon(z)\partial\phi - \bar{\epsilon}(\bar{z})\bar{\partial}\phi$, 生成元 $l_n = -z^{n+1}\partial_z$, $\bar{l}_n = -\bar{z}^{n+1}\partial_{\bar{z}}$

对易: $[l_m, l_n] = (m-n)l_{m+n}$

$[\bar{l}_m, \bar{l}_n] = (m-n)\bar{l}_{m+n}$

$[l_m, \bar{l}_n] = 0$

子代数
 $\Rightarrow$

$l_1 = -z^2\partial_z$ SCT for $\mathbb{C}$

$l_0 = -z\partial_z$ dilation, rotation

$l_{-1} = -\partial_z$ translation for $\mathbb{C}$

保物理性变换:

$\Rightarrow l_n + \bar{l}_n, i(l_n - \bar{l}_n)$

$\downarrow$
 $l_0 + \bar{l}_0$ 为 dilation
 $i(l_0 - \bar{l}_0)$ 为 rotation

初等场: 标度 Δ , 自旋 S ,

全纯共形维 $h = \frac{1}{2}(\Delta + S) \Rightarrow L_0 = h - z\partial, \bar{L}_0 = -\bar{h} - \bar{z}\bar{\partial}$

反全纯共形维 $\bar{h} = \frac{1}{2}(\Delta - S)$ 故 $\phi'(w, \bar{w}) = \left(\frac{dw}{dz}\right)^h \left(\frac{d\bar{w}}{d\bar{z}}\right)^{\bar{h}} \phi(z, \bar{z})$

无穷小场变: $w = z + \epsilon(z), \bar{w} = \bar{z} + \bar{\epsilon}(\bar{z})$, $\delta_{\epsilon, \bar{\epsilon}} \phi = \phi'(z, \bar{z}) - \phi(z, \bar{z}) = \phi'(w, \bar{w}) - \epsilon\partial\phi - \bar{\epsilon}\bar{\partial}\phi - \phi = (1+\partial\epsilon)^h (1+\bar{\partial}\bar{\epsilon})^{\bar{h}} \phi - \epsilon\partial\phi - \bar{\epsilon}\bar{\partial}\phi - \phi$
 $= -(h\partial\epsilon\phi + \epsilon\partial\phi) - (\bar{h}\bar{\partial}\bar{\epsilon}\phi + \bar{\epsilon}\bar{\partial}\phi)$

关联函数: 对 $h, \bar{h}$, 全纯坐标 $w, \bar{w}$

$$\downarrow \langle \phi_1(w_1, \bar{w}_1) \dots \phi_n(w_n, \bar{w}_n) \rangle = \prod_{i=1}^n \left(\frac{dw_i}{dz_i}\right)^{-h_i} \left(\frac{d\bar{w}_i}{d\bar{z}_i}\right)^{-\bar{h}_i} \langle \phi_1(z_1, \bar{z}_1) \dots \phi_n(z_n, \bar{z}_n) \rangle$$

$$2\text{点}: \langle \phi_1(z_1, \bar{z}_1) \phi_2(z_2, \bar{z}_2) \rangle = \frac{C_{12}}{(z_1 - z_2)^{2h} (\bar{z}_1 - \bar{z}_2)^{2\bar{h}}}, \text{ if } h_1 = h_2 = h, \bar{h}_1 = \bar{h}_2 = \bar{h}$$

旋转不变: $|x_{ij}| = (z_{ij}\bar{z}_{ij})^{\frac{1}{2}}$, $\langle X \rangle = f(|x_{ij}|)$, 故 $h = \bar{h} \Rightarrow \sum S_i = 0$. 总自旋为0

$$3\text{点} \langle \phi_1(z_1, \bar{z}_1) \phi_2(z_2, \bar{z}_2) \phi_3(z_3, \bar{z}_3) \rangle = C_{123} \frac{1}{z_{12}^{h_1+h_2-h_3} z_{23}^{h_2+h_3-h_1} z_{13}^{h_1+h_3-h_2}} \frac{1}{\bar{z}_{12}^{\bar{h}_1+\bar{h}_2-\bar{h}_3} \bar{z}_{23}^{\bar{h}_2+\bar{h}_3-\bar{h}_1} \bar{z}_{13}^{\bar{h}_1+\bar{h}_3-\bar{h}_2}}$$

旋转不变: 同上, $\sum S_i = 0$?

$$4\text{点}: \text{非谐比形式: } \eta = \frac{z_{12}z_{34}}{z_{13}z_{24}} \rightarrow \frac{z_{14}z_{23}}{z_{13}z_{24}} = 1-\eta, \frac{z_{12}z_{34}}{z_{14}z_{23}} = \frac{\eta}{1-\eta}$$

$$\text{故 } \langle \phi_1 \dots \phi_4 \rangle = f(\eta, \bar{\eta}) \prod_{i,j} z_{ij}^{\frac{1}{2}h_i-h_j} \bar{z}_{ij}^{\frac{1}{2}\bar{h}_i-\bar{h}_j}$$

全局共形: 3自由: 使 $z_1, z_2, z_3 \rightarrow$ 任三点

取 $z_1=1, z_2=\infty, z_3=0 \Rightarrow \eta = -z_4$, 故一般4点函数关于极点 ?

2. 共形Wald

$$\left\{ \begin{array}{l} \text{平移: } \frac{\partial}{\partial x^\mu} \langle T^\mu_\nu(x) X \rangle = - \sum_{i=1}^n \delta(x-x_i) \frac{\partial}{\partial x^\nu} \langle X \rangle \\ \text{转动: } \epsilon_{\mu\nu} \langle T^{\mu\nu}(x) X \rangle = -i \sum_{i=1}^n s_i \delta(x-x_i) \langle X \rangle \quad s_i \text{ 为自旋, } \Delta_i \text{ 为标度} \\ \text{标度: } \langle T^\mu_\mu(x) X \rangle = - \sum \delta(x-x_i) \Delta_i \langle X \rangle \end{array} \right.$$

⊂'坐标下:

$$\text{Gauss: } \int_M d^2x \partial_\mu F^\mu = \int_{\partial M} d\xi_\mu F^\mu, \text{ 有: } d\xi_\mu = \epsilon_{\mu\rho} ds^\rho$$



$$\text{故: } \int_M d^2x \partial_\mu F^\mu = \int_{\partial M} F^\mu d\xi_\mu = \frac{i}{2} \int_{\partial M} (-dz F^{\bar{z}} + d\bar{z} F^z),$$

$$\hookrightarrow \int_M d^2x \delta(x) f(z) = f(0) = \frac{1}{2\pi i} \int_{\partial M} dz \frac{f(z)}{z} = \frac{1}{2\pi} \int_M d^2x \bar{\partial} \left(\frac{f(z)}{z} \right) = \frac{1}{2\pi} \int_M d^2x f(z) \bar{\partial} \frac{1}{z}$$

$$\text{取 } F^{\bar{z}} = \frac{f(z)}{z}, F^z = 0$$

$$\text{即: } \delta(x) = \frac{1}{2\pi} \bar{\partial} \frac{1}{z} = \frac{1}{2\pi} \partial \frac{1}{\bar{z}}$$

$$\text{代入: } 2\pi \partial \langle T_{zz} X \rangle + 2\pi \bar{\partial} \langle T_{z\bar{z}} X \rangle = - \sum_{i=1}^n \bar{\partial} \frac{1}{z-w_i} \partial w_i \langle X \rangle$$

$$2\pi \partial \langle T_{\bar{z}\bar{z}} X \rangle + 2\pi \bar{\partial} \langle T_{z\bar{z}} X \rangle = - \sum_{i=1}^n \partial \frac{1}{z-w_i} \bar{\partial} w_i \langle X \rangle$$

$$\left. \begin{array}{l} 2\langle T_{z\bar{z}} X \rangle + 2\langle T_{\bar{z}z} X \rangle = - \sum_{i=1}^n \delta(x-x_i) \Delta_i \langle X \rangle \\ -2\langle T_{z\bar{z}} X \rangle + 2\langle T_{\bar{z}z} X \rangle = - \sum_{i=1}^n \delta(x-x_i) s_i \langle X \rangle \end{array} \right\} \begin{array}{l} \langle T_{\bar{z}z} X \rangle = \frac{1}{2\pi} \sum_{i=1}^n \bar{\partial} \frac{1}{z-w_i} h_i \langle X \rangle \rightarrow -2\pi \bar{\partial} \langle T_{z\bar{z}} X \rangle = \bar{\partial}^2 \sum \frac{h_i}{z-w_i} \langle X \rangle = \bar{\partial} \sum \frac{-h_i}{(z-w_i)^2} \langle X \rangle \\ \langle T_{z\bar{z}} X \rangle = \frac{1}{2\pi} \sum_{i=1}^n \partial \frac{1}{z-w_i} \bar{h}_i \langle X \rangle \rightarrow -2\pi \partial \langle T_{\bar{z}z} X \rangle = \partial^2 \sum \frac{-\bar{h}_i}{(\bar{z}-\bar{w}_i)^2} \langle X \rangle \end{array}$$

$$\text{上两式联立: } T_{(z,\bar{z})} = -2\pi T_{\bar{z}z} \quad \bar{\partial} \left[\langle T_{(z,\bar{z})} X \rangle - \sum_{i=1}^n \left(\frac{1}{z-w_i} \partial w_i \langle X \rangle + \frac{h_i}{(z-w_i)^2} \langle X \rangle \right) \right] = 0 \rightarrow \text{积分余一个全纯函数 } \partial f(z) = 0$$

$$\bar{T}_{(z,\bar{z})} = -2\pi T_{z\bar{z}} \quad \partial \left[\langle \bar{T}_{(z,\bar{z})} X \rangle - \sum_{i=1}^n \left(\frac{1}{\bar{z}-\bar{w}_i} \bar{\partial} \bar{w}_i \langle X \rangle + \frac{\bar{h}_i}{(\bar{z}-\bar{w}_i)^2} \langle X \rangle \right) \right] = 0 \quad \text{且: 内为全纯/反全纯} \rightarrow T(z), \bar{T}(\bar{z})$$

共形关系

$$\text{共形小变换: } x' = x + \epsilon, \quad \partial_\mu (E_\nu T^{\mu\nu}) = E_\nu \partial_\mu T^{\mu\nu} + \frac{1}{2} (\partial_\mu E_\nu + \partial_\nu E_\mu) T^{\mu\nu} + \frac{1}{2} (\partial_\mu E_\nu - \partial_\nu E_\mu) T^{\mu\nu} \\ = E_\nu \partial_\mu T^{\mu\nu} + \frac{1}{2} (\partial_\rho E^\rho) \eta_{\mu\nu} T^{\mu\nu} + \frac{1}{2} \epsilon^{\alpha\beta} \partial_\alpha E_\beta \epsilon_{\mu\nu} T^{\mu\nu}$$

$$\downarrow \\ \delta \epsilon \langle X \rangle = \int_M d^2x \partial_\mu \langle T^{\mu\nu}(x) E_\nu(x) X \rangle$$

$$\text{取 } \langle T^{\mu\nu} \epsilon_\nu X \rangle = F^\mu \rightarrow \text{Gauss: } \delta \epsilon \cdot \bar{\epsilon} \langle X \rangle = \frac{i}{2} \int_{\partial M} (-dz \langle T^{\bar{z}\bar{z}} \epsilon_{\bar{z}} X \rangle + d\bar{z} \langle T^{zz} \epsilon_z X \rangle)$$

↓

∂M 不经过 X 中的点, 所有 $\langle T^{\mu\nu} X \rangle = -\sum \delta(x-x_i) \Delta_i X \rightarrow 0$, ϵ^μ

$$\text{共形Wald恒等式: } \delta \epsilon \bar{\epsilon} \langle X \rangle = \frac{-1}{2\pi i} \oint_C dz \epsilon(z) \langle T(z) X \rangle + \frac{1}{2\pi i} \oint_C d\bar{z} \bar{\epsilon}(\bar{z}) \langle \bar{T}(\bar{z}) X \rangle, \quad C \text{ 围所有 } (w_i, \bar{w}_i), \quad X \text{ 可为局域场}$$

$$X \text{ 为初场: } \delta \epsilon \langle X \rangle = -\sum_i (\epsilon c w_i \partial w_i + \partial \epsilon(w_i) h_i) \langle X \rangle, \quad \text{SL}(2, \mathbb{C}): f(z) = \frac{(1+\alpha)z + \beta}{\gamma z + 1 - \alpha} : \epsilon(z) = \beta + 2\alpha z - \gamma z^2 \quad \text{对 } \forall \alpha, \beta, \gamma \text{ 不变:}$$

$$\hookrightarrow \text{对场: } \delta_\epsilon \phi = -\epsilon \partial \phi - h \phi \partial \epsilon$$

$$\hookrightarrow \sum \partial w_i \langle \phi_1 \cdots \phi_n \rangle = 0$$

$$\sum (w_i \partial_i + h_i) \langle \phi_1 \cdots \phi_n \rangle = 0$$

$$\sum (w_i \partial_i + 2w_i h_i) \langle \phi_1 \cdots \phi_n \rangle = 0$$